

Introduction

Of the many things Intune is good at, you will not find reporting listed. This is because the design phase at Microsoft, like all large corporate IT projects, did not take compliance monitoring and audit requirements seriously. They have an extensive list of configuration settings and compliance policies, but getting that data in a meaningful form from the Intune console can be a challenge.

There is a common perception that console snapshots can provide enough evidence and that if there is a requirement for better or more detailed evidence that you can push back to the auditor and they will go away. In fact, the opposite always happens, an audit finding gets opened and you now have no choice but to find a way to get that evidence.

Key drivers

“Trust but verify” (Zero Trust)

A good example is Bitlocker where you can require bitlocker encryption on all windows devices, see in the console that you have X% compliant and non-compliant devices, but can't prove why they are compliant or not. It can be difficult to determine if the drive protector is on, if the bitlocker service is running. These are some examples of the many other Key Compliance Indicators (KCI's) an auditor might ask for evidence of.

If you have a “trust but verify” approach, then you need to collect and provide details that support the X% compliant calculation. You never let the fox watch the henhouse.

Impact on Source and vice versa (Throttling and Security)

Your reporting using Graph API has an impact felt across your tenant. If you are using SharePoint, One Drive and Teams, are onboarding and offboarding devices, and generally just using the product portals, all this traffic uses the Graph API at one point or another.

Each team must work to have a minimal impact on Graph API usage due to the tenant restriction of 130,000 requests per 10 seconds. This may not sound significant, but when you have 70,000 users and 300,000 devices, you can quickly hit that throttling wall querying configuration settings and device compliance status, especially if other support teams are performing a migration or permissions audit.

The best solution is to replicate the data to a “sandbox” once to isolate the golden source from the impact of your team’s activities and then your reporting analysts can perform as many queries as they want without impacting other teams’ functionality.

This also has the added benefit of increasing your security posture in that none of the data consumers have any permissions in the Graph API. All you need to do is make sure the replication process uses an App Registration, replicate the data to the “sandbox” and control access to the “sandbox.”

Data volumes and Time constraints for report production (ETL Processes)

I saw a demonstration using Power BI to produce a compliance dashboard. It was pretty responsive in the lab with 4 devices. Scaling it up showed its limitations. Performing ETL processes using Power Query is not very efficient. Filtering data at the source is an option, collecting just what you need, but you may not get all your dashboards updated in time for the data consumers delivery requirements.

As an example, each report may have several requirements in common with other reports. Most of my reports contain inventory data such as device name, serial, location (to the city) as well as business information for device ownership. If this process takes one hour to create, and is used by 8 reports, then by saving this common data and re-using it for the other reports can save 7 hours of effort overall to produce the 8 reports.

The SQL “sandbox” ingesting the same filtered data can normalize and index the result data for speed, combine repeated actions to create derived sources for re-use, and dimension the data to reduce required storage thus optimizing the data pipeline to produce results faster than Power Query.

Audit and enterprise visibility (SDLC and segregation of environments)

Auditors and Cybersecurity groups often demand compliance statistics for all machines in the environment. This includes the DEV and UAT tenants as well as the Production tenant and requires combining the data across these sources to provide a unified view of device coverage.

For instance, the agent for an intrusion detection system or vulnerability scanning needs to be deployed to every system regardless of environment and operating system. Without a method to verify their proper deployment and operation, you could be exposed and never know.

Historical evidence retention requirements (Audit requirement)

Data in Azure and Intune have a limited lifetime, often 30 days. Auditors may require that the compliance data you generate persist for up to a year. Having a “sandbox” allows you to meet that requirement easily.

Components

PostGres, the “sandbox”

App Registration (Service Principal)

Data Collection Components

ETL Processes

Process Orchestration

PGAdmin?

REST as a data source

